

Long-term effects of a reduced fat diet intervention on cardiovascular disease risk factors in individuals with glucose intolerance

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Abstract

The long-term effects on cardiovascular disease risk factors of a reduced fat (RF), ad libitum diet were compared with usual diet (control, CD) in glucose intolerance individuals.

Participants were 136 adults aged ≥ 40 years with 'glucose intolerance' (2 h blood glucose 7–11.0 mmol/l) detected at a Diabetes Survey who completed at 1 year intervention study of reduced fat, ad libitum diet versus usual diet. They were re-assessed at 2, 3 and 5 years. Main outcome measures were blood pressure, serum concentrations of total cholesterol, HDL and LDL cholesterol, total cholesterol:HDL ratio, triglycerides and body weight.

The reduced fat diet lowered total cholesterol ($P < 0.01$), LDL cholesterol ($P \leq 0.05$), total cholesterol:HDL ratio ($P \leq 0.05$), body weight ($P < 0.01$) and systolic blood pressure ($P \leq 0.05$) initially and diastolic blood pressure ($P < 0.01$) long-term. No significant changes occurred in HDL cholesterol or triglycerides. In the more compliant 50% of the intervention group, systolic and diastolic blood pressure levels and body weight were lower at 1, 2 and 3 years ($P < 0.05$).

It was concluded that a reduced fat ad libitum diet has short-term benefits for cholesterol, body weight and systolic blood pressure and long-term benefits for diastolic blood pressure without significantly effecting HDL cholesterol and triglycerides despite participants regaining their lost weight.

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1. Introduction

People with impaired glucose tolerance (IGT) are known to be at high risk of developing type 2 diabetes

[1], and have an independent risk of atherosclerosis nearly three times that of people with normal glucose tolerance (NGT), after 5 years of follow up [2]. The rate at which people with IGT die from cardiovascular disease is also higher than normoglycaemic controls [3,4].

High dietary fat intake has been consistently linked to the development of IGT and diabetes [5,6]. A high fat diet promotes passive over-consumption of total

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energy [7] and weight gain is a critical factor in both the development of IGT [8] and its progression to diabetes [9], increasing diabetes risk approximately, 2–12-fold [10].

There is substantial current debate about the impact of reducing total dietary fat. Proponents of ad libitum reduced fat diets maintain that they promote weight loss [11–13]. Weight loss is known to reduce blood pressure, serum triglycerides, total and LDL cholesterol, blood glucose and increase HDL cholesterol [14], all of which are important for people with IGT [15].

However, metabolic studies with isoenergetic substitution of carbohydrate for fat have produced mixed results on blood lipids. One such study in people with IGT reported adverse changes in triglyceride and HDL cholesterol concentrations [16], and another similar study in people with type 2 diabetes reported persistent elevation of triglyceride levels [17]. However, others have found no adverse effects with isoenergetic studies in obese and/or type 2 diabetes participants [18] or healthy normal weight volunteers [19].

Short-term dietary manipulations that precisely replace energy from fat with carbohydrate may not provide appropriate evidence for recommendations on reducing fat for health gains in free living individuals or populations. Reducing total fat intake whilst maintaining an otherwise ad libitum diet without attempting to replace energy removed, does reduce energy intake [20] and either decreases weight in the overweight or prevents weight gain in normal weight people [21]. A previous report of our glucose intolerant group found that the energy deficit from reduced fat eating was not replaced with carbohydrate [22].

We have previously reported on the 5 year effects of a 1 year intervention for reduced fat eating on blood glucose and glucose tolerance status [23]. A significant positive effect of diet was found on 2 h glucose over a 5 year period ($P < 0.0001$) and on glucose tolerance status at 1 year ($P = 0.015$). Also the people who were most compliant with the program showed long-term reductions in both fasting ($P = 0.041$) and 2 h ($P = 0.024$) blood glucose levels. We now examine the long-term impact of the reduced fat ad libitum intervention on other known cardiovascular disease risk factors.

2. Materials and methods

2.1. Participants

Participants were selected from a previous Workforce Diabetes Survey [24] in which glucose tolerance tests were performed in 4833 workers aged 40 years and over, from 41 work sites around Auckland between 1988 and 1990. One-hundred and forty-five (3%) of those surveyed were classified as having impaired glucose tolerance by the 1985 WHO criteria (2 h blood glucose 7.8–11.0 mmol/l) [25] and a further 114 (2%) had high normal blood glucose concentrations (7.0–7.8 mmol/l). In this combined 'glucose intolerant' group (i.e. 2 h blood glucose 7.0–11.0 mmol/l, $n = 276$), 237 could be contacted (approximately, 2 years after the original survey), 176 entered the intervention and 136 (77%) completed the 1 year intervention period. Of these, 136 participants, 104 (76%) were reassessed at 2 years, 99 (73%) at 3 years and 112 (82%) at 5 years.

Prior to the intervention, participants underwent a repeat oral glucose tolerance test to redefine their glucose tolerance status, which in many cases had changed since their original test in the Workforce Diabetes Survey 2 years previously [23]. They were then individually randomised to either an intervention group that was asked to consume a reduced-fat (RF), but otherwise ad libitum diet, or a control diet (CD) group that continued with their usual diet. An exception to this individual randomisation was made at one work-site where all six participants were Pacific Island's women who worked closely together. They were all assigned to the RF group because individual randomisation was impractical. All those found to have diabetes on re-testing were referred to their general practitioners for management, but were still randomised for the study.

2.2. Dietary intervention

Participants in the RF group entered a 1 year structured programme aimed solely at reducing the total amount of fat in their diet. The programme involved education on reducing dietary fat intake, personal goal setting and self-monitoring (food diaries) as previously described [26]. Compliance with the programme was assessed by attendance rate at monthly meetings

and completion of diet diaries (compliance score = mean of the percentage of meetings attended and the percentage of diaries filled). Those with compliance scores above the median for the group were classified as ‘compliers’ for the analysis.

2.3. Measurements

Before randomisation and at the end of the study participants in both groups completed a comprehensive 3 day food diary. Food diaries were analysed for macronutrients, alcohol, fibre, cholesterol and calcium using Nutritionist III [27] software with the nutrient values of about 600 New Zealand foods added. At 5 years, dietary intake over the past 3 months was estimated using a previously validated self-administered 142-item food frequency questionnaire supplemented with New Zealand foods [28].

Participants provided fasting venous blood samples that were analysed for serum total cholesterol, HDL and triglycerides using enzymatic methods. LDL cholesterol concentrations were calculated using the method of Friedewald [29] provided the total triglyceride concentrations were less than 4.6 mmol/l. After a rest of 5 min or more blood pressure was measured twice in the sitting position (right arm) with a Hawksley random zero sphygmomanometer at baseline, 1 and 2 years. An average systolic and average diastolic blood pressure was calculated from these two measurements if they were within 10 mmHg of each other otherwise a third measurement was taken and the mean of all three was used. At 1 and 3 years, an electronic measurement of blood pressure was also done using an Omron Hem-400C (Omron Tateisi, Tokyo). An Omron Hem-706 oscillometric blood pressure pulse monitor was used at 5 years. A standard 75 g oral glucose tolerance test (fasting and 2 h samples) was performed at each of these points. Diabetes, IGT and impaired fasting glycaemia were defined using the revised WHO criteria and classification of diabetes mellitus [30].

2.4. Statistical analyses

Mean (S.E.M.) values, adjusted for age, sex, ethnicity and baseline measurements, were calculated using SAS PROC MIXED (SAS Institute, Cary, NC) [31]. The repeated-measures model included time

since baseline for each of the time points, treatment group, and the time by treatment interaction, with an unstructured variance–covariance matrix to model the correlations between time points. The Satterthwaite option was used to approximate the denominator degrees of freedom for the tests of fixed effects. A linear model was assumed from time 0–5 years. Categorical variables were compared using the χ^2 -test and Student’s *t*-test was used for continuous variables. All statistical analyses were carried out using SAS [31]. An intention-to-treat analysis was not carried out in participants who dropped out before the end of the study because if their last observation had been carried forward, the differences between the two groups would have been artificially maintained over the follow-up period. Carrying the last observation forward assumes that assigning a no-change status is a conservative analysis, which is not the case, because weight change over time was non-linear.

3. Results

3.1. Participants

The baseline characteristics of participants who completed the 1 year intervention are presented in Table 1. There were no significant differences found between the groups except that there were more smokers in the reduced fat diet group ($P = 0.02$). Forty people did not complete the 1 year intervention (four died, one became pregnant, seven developed serious illness, four moved out of Auckland and 24 dropped out of the study). More Maori and Pacific Islands participants did not complete the intervention period ($P = 0.010$), but otherwise the characteristics of the completers and non-completers were not significantly different. The baseline characteristics of the participants who were lost to 5-year follow up ($n = 24$) were not significantly different from the other participants except they were older (53 years 5 months versus 49 years 9 months, $P = 0.029$).

3.2. Dietary intake

The daily macronutrient intakes, fat fractions (saturated, monounsaturated and polyunsaturated), alcohol, fibre, cholesterol and calcium intake determined

Table 1

Baseline characteristics of participants who completed the 1 year intervention

	Reduced fat diet (<i>n</i> = 66)	Control diet (<i>n</i> = 70)
Age (years) (S.E.M.)	52.5 (0.8)	52.0 (0.8)
Gender M:F	45:21	56:14
Ethnicity		
European (<i>n</i> %)	44 (66.7)	53 (75.7)
Maori (<i>n</i> %)	7 (10.6)	5 (7.1)
Pacific islander (<i>n</i> %)	13 (19.7)	9 (12.9)
Other (<i>n</i> %)	2 (3.0)	3 (4.3)
Smoking status		
Non-smoker (<i>n</i> %)	51 (77)	64 (91) ^a
Current smoker (<i>n</i> %)	15 (23)	6 (9)
Anthropometry		
Weight (kg) (S.E.M.)	85.30 (1.84)	84.83 (1.79)
Body mass index (kg/m ²) (S.E.M.)	29.29 (0.57)	29.12 (0.56)
Waist circumference (cm) (S.E.M.)	100.53 (1.45)	101.58 (1.40)
Waist to hip ratio (S.E.M.)	0.942 (0.009)	0.957 (0.009)
Lipid tests		
Cholesterol (mmol/l) (S.E.M.)	6.07 (0.12)	5.96 (0.12)
HDL cholesterol (mmol/l) (S.E.M.)	1.14 (0.04)	1.12 (0.04)
LDL cholesterol (mmol/l) (S.E.M.)	4.08 (0.11)	3.96 (0.11)
Triglyceride (mmol/l) (tolerance factor)	1.67 (1.14)	1.73 (1.13)
Blood pressure		
Systolic (mmHg) (S.E.M.)	140.76 (2.67)	146.30 (2.59)
Diastolic (mmHg) (S.E.M.)	89.58 (1.56)	89.99 (1.52)
Glucose tolerance status		
Normal (<i>n</i> %)	24 (36.4)	23 (32.9)
Impaired fasting glucose (<i>n</i> %)	11 (16.6)	15 (21.4)
Impaired glucose tolerance (<i>n</i> %)	10 (15.1)	9 (12.9)
Diabetes (<i>n</i> %)	21 (31.8)	23 (32.9)

^a Significantly different *P* = 0.02.

from 3-day food diaries are presented in Table 2. At the end of the 1 year intervention, full dietary data were available from 110 participants (81%). Over the year, the RF group reduced their total energy intake by 1.5 MJ per day and their fat from 35 to 26% of energy intake. The reduction was similar for the 3 fat types measured (saturated 63%, monounsaturated

66%, polyunsaturated 64% of baseline fat intake). The P:M:S ratio at baseline (1:2.2:2.6) was similar to that after 1 year of intervention (1:2.2:2.5). Although there was no change in absolute carbohydrate intake, the RF group significantly increased their proportional energy from carbohydrate (46–54%) and protein (17–18%), and their fibre intake (2.3–2.7 g/MJ) compared with controls. Saturated fat intake dropped by over 14 g per day (*P* = 0.0002).

The food frequency questionnaire carried out at 5 years revealed no differences between the RF and CD groups in macronutrient intake (when expressed as a percentage of energy) and no differences in cholesterol and fibre intake expressed as g/MJ. The RF group had a higher intake of calcium (99 mg/MJ versus 85 mg/MJ, *P* = 0.0163) than controls.

3.3. Biochemical and physiological measurements

The mean changes in serum lipids (total cholesterol, HDL, LDL, total cholesterol:HDL ratio and triglycerides), body weight, systolic and diastolic blood pressure observed at 0.5, 1, 2, 3 and 5 years are presented in Table 3, adjusted for age, gender, ethnicity and baseline measurements. In the RF group, total, LDL cholesterol and total cholesterol:HDL ratio were significantly lower at 0.5 years and overall (*P* < 0.0001, *P* = 0.0006 and *P* = 0.008, respectively) during the 5 year period. HDL cholesterol levels and triglycerides in the RF group were not different at any time point between the groups, but did show non-parallel differences (*P* = 0.02, *P* = 0.01, respectively). The RF group lost weight during the intervention and this was sustained short term. They had a significantly lower systolic blood pressure at 0.5 years and overall (*P* = 0.019) and lower diastolic blood pressure at 1, 2 and 5 years, and overall (*P* = 0.056). A comparison of the Hawksely random zero sphygmomanometer and the Omron electronic monitor in a larger group from the same Workforce Diabetes Survey (*n* = 234) revealed the electronic device read 1.74 (S.E. = 0.85) mmHg higher for systolic blood pressure and 1.26 (S.E. = 0.59) mmHg higher for diastolic blood pressure than the random zero measurements (*P* = 0.04 for both).

At baseline, 12 (18%) participants in the low fat diet group and 19 (27%) participants in the control diet group were taking blood pressure medications. At 0.5 and 1 year, the numbers taking blood pressure

Table 2

Mean daily energy, fat and fibre intakes (S.E.M.) determined by 3 day diet diaries adjusted for age, sex and ethnicity

	Reduced fat diet (<i>n</i> = 49)			Control diet (<i>n</i> = 61)			Changes (<i>P</i>) ^a
	Baseline	1 year	Change	Baseline	1 year	Change	
Energy (MJ)	9.4 (0.4)	7.9 (0.4)	−1.5 (0.4)	9.7 (0.3)	9.5 (0.4)	−0.3 (0.4)	0.0258
Carbohydrate (g)	256 (10)	256 (11)	0 (12)	255 (9)	246 (10)	−10 (11)	NS
(%Energy)	46.3 (1.3)	54.2 (1.5)	7.9 (1.4)	45.0 (1.1)	45.8 (1.4)	0.9 (1.2)	<0.0004
Protein (g)	92.4 (3.9)	85.5 (5.1)	−6.9 (4.4)	95.2 (3.5)	93.5 (4.6)	−1.7 (3.9)	NS
(%Energy)	16.7 (0.5)	18.4 (0.5)	1.7 (0.6)	16.8 (0.4)	16.6 (0.5)	−0.2 (0.5)	0.0207
Fat (g)	87.6 (4.8)	54.5 (5.0)	−33.0 (5.0)	95.3 (4.3)	88.1 (4.5)	−7.3 (4.5)	0.0002
(%Energy)	34.5 (0.9)	26.1 (1.1)	−8.4 (1.2)	36.2 (0.8)	33.6 (1.0)	−2.6 (1.1)	0.0006
Alcohol (g)	15.5 (3.7)	10.9 (3.9)	−4.6 (3.4)	16.8 (3.3)	21.3 (3.5)	4.6 (3.0)	0.0491
(%Energy)	4.6 (0.9)	3.6 (1.0)	−1.0 (0.8)	4.3 (0.8)	5.7 (0.9)	1.3 (0.7)	0.0331
Saturated fat (g)	35.3 (2.3)	20.9 (2.3)	−14.4 (2.3)	37.7 (2.0)	35.4 (2.1)	−2.3 (2.1)	0.0002
(%Energy)	13.8 (0.5)	10.0 (0.6)	−3.8 (0.6)	14.2 (0.5)	13.4 (0.6)	−0.8 (0.5)	0.0004
Monounsaturated (g)	30.0 (1.8)	18.4 (1.8)	−11.6 (1.9)	33.6 (1.6)	31.2 (1.6)	−2.4 (1.7)	0.0006
(%Energy)	11.7 (0.4)	8.9 (0.4)	−2.9 (0.5)	12.8 (0.3)	11.8 (0.4)	−1.0 (0.5)	0.0086
Polyunsaturated (g)	13.3 (0.8)	8.5 (0.7)	−4.9 (0.8)	13.2 (0.7)	12.0 (0.6)	−1.3 (0.7)	0.0017
(%Energy)	5.4 (0.2)	4.0 (0.2)	−1.4 (0.3)	5.2 (0.2)	4.8 (0.2)	−0.4 (0.2)	0.0067
Fibre (g)	21.1 (1.1)	20.8 (1.1)	0.4 (1.3)	19.2 (1.0)	18.1 (1.0)	−1.1 (1.1)	NS
(g/MJ)	2.3 (0.1)	2.7 (0.1)	−0.3 (0.1)	2.1 (0.1)	2.1 (0.1)	0.1 (0.1)	0.0436
Cholesterol (g)	394 (30)	308 (36)	−86 (35)	417 (27)	419 (32)	2 (31)	NS
Calcium (g)	665 (37)	627 (41)	−38 (45)	666 (33)	623 (36)	−43 (40)	NS

^a Difference in changes between reduced fat and control diet groups.

medications in the LF group dropped to 2 (3.1) and 6 (9.2%), respectively and in the CD group dropped to 5 (7.1) and 7(10%), respectively. None of these differences were significant. There were no significant differences found between groups in their cholesterol lowering medications or changes to these medications at 0.5, 1, 2 or 3 years.

3.4. Effects with greater compliance

There were no differences in gender or glucose tolerance status between those participants in the reduced fat group classified as ‘compliers’ or ‘non-compliers’. ‘Compliers’ were more likely to be European ($P = 0.001$). Fig. 1 shows the effects (actual, measurements and unadjusted) of compliance on blood lipids, body weight and blood pressure. The ‘compliers’ had significantly lower body weight at every time point up to 3 years, and systolic and diastolic blood pressure at 1, 2 and 3 years compared with the CD group. However,

greater compliance showed no effect on total cholesterol, HDL cholesterol, LDL cholesterol, total cholesterol:HDL ratio or triglyceride levels.

4. Discussion

The medium term effects (1 year) of the reduced fat education program were a reduction in fat intake with no compensatory increase in absolute intake of carbohydrate. This resulted in a reduction in total energy intake (84% from fat reduction), weight loss and no change in the P:M:S ratio. Proportionally, more energy was obtained from carbohydrates-, protein- and fibre-rich foods. These changes resulted in overall lower total cholesterol, LDL cholesterol and blood pressure during the 5 year period, but particularly in the first 6 months. The RF group had significantly lower diastolic blood pressure at 1, 2 and 5 years compared to controls, despite participants regaining

Table 3

Mean changes in risk factor measurements (S.E.M.) from baseline over 5 years in reduced fat (RF, $n = 66$) and control diet (CD, $n = 70$) groups adjusted for age, gender, ethnicity and baseline measurements

RF/CD (N)		Intervention period					Overall effect of diet*
		66/70 ^c		47/57 ^c	48/51 ^c	51/52 ^c	
		(0.5 years)	(1 years)	(2 years)	(3 years)	(5 years)	
Total cholesterol (mmol/l)	RF	−0.37 (0.07) ^b	−0.15 (0.09)	−0.06 (0.11)	−0.12 (0.11)	−0.20 (0.11)	<0.0001
	CD	−0.07 (0.18)	−0.05 (0.17)	0.14 (0.19)	−0.06 (0.18)	−0.15 (0.18)	
LDL cholesterol (mmol/l)	RF	−0.26 (0.09) ^a	−0.18 (0.09)	−0.07 (0.09)	−0.08 (0.10)	−0.32 (0.09)	0.0006
	CD	0.01 (0.15)	−0.02 (0.15)	0.01 (0.17)	−0.08 (0.15)	−0.16 (0.16)	
HDL cholesterol (mmol/l)	RF	0.01 (0.02)	−0.02 (0.02)	0.08 (0.04)	−0.03 (0.03)	0.01 (0.02)	0.0213
	CD	−0.03 (0.05)	0.01 (0.05)	0.06 (0.05)	−0.01 (0.05)	0.06 (0.05)	
Total cholesterol:HDL ratio	RF	−0.37 (0.10) ^a	−0.10 (0.14)	−0.44 (0.13)	−0.17 (0.13)	−0.34 (0.14)	0.0008
	CD	−0.01 (0.26)	−0.11 (0.26)	−0.23 (0.28)	−0.12 (0.27)	−0.53 (0.24)	
Weight (kg)	RF	−2.97 (0.54) ^b	−3.32 (0.68) ^b	−3.15 (0.78) ^b	−1.60 (0.78) ^b	1.06 (0.64)	<0.0001
	CD	−0.08 (0.43)	0.59 (1.61)	1.06 (0.46)	2.13 (0.70)	1.26 (0.68)	
Triglycerides (mmol/l)	RF	−0.12 (0.12)	0.03 (0.11)	−0.11 (0.15)	−0.02 (0.14)	0.37 (0.10)	0.0137
	CD	−0.01 (0.23)	0.07 (0.21)	0.01 (0.22)	0.25 (0.27)	0.12 (0.22)	
Systolic blood pressure (mmHg)	RF	−4.31 (1.96) ^a	−4.10 (1.99)	−2.99 (2.19)	−3.63 (1.60)	−3.54 (2.48)	0.0185
	CD	−1.99 (2.97)	3.37 (3.21)	2.56 (3.18)	2.81 (3.10)	1.31 (3.38)	
Diastolic blood pressure (mmHg)	RF	−1.31 (1.54)	−1.73 (1.51) ^b	−6.10 (1.49) ^b	−5.78 (1.27)	−7.16 (1.68) ^a	<0.0001
	CD	−1.57 (2.10)	2.11 (2.09)	−1.87 (2.00)	−2.47 (2.05)	−4.20 (1.92)	

* Effect of diet assessed by the interaction term group $\times$ time. Significant difference between RF and CD groups at that time point.

^a $0.01 = P = 0.05$.

^b $P < 0.0$.

^c RF/CD.

weight they had lost and a more conservative blood pressure reduction obtained with the electronic device at 5 years. No significant changes in HDL cholesterol and triglycerides were observed at any time point, but overall HDL and triglycerides showed non-parallel differences over time. More compliant individuals had consistently lower systolic and diastolic blood pressure after the intervention and up to 2 years later compared to the CD group.

Education to reduce dietary fat for health reasons was successful in achieving moderate changes to the proportions that carbohydrate and fat contribute to total energy (from 46–54 and 35–26%, respectively). At the same time, an unintentional energy deficit resulted in an initial weight loss in this group. As men who gained >10% weight from age 20 years to middle-age significantly increased their chances of developing hypertension, dyslipidaemia and hyperinsulinaemia (19, 11 and 38%, respectively) [8], there are considerable benefits to be gained from weight loss alone in an invited workforce population.

Although the limitations of using different dietary methods have to be taken into account, the lack of

significant differences in food intake, other than calcium, at 5 years between the two groups would suggest that most dietary changes, whilst significant when supported by health professional follow-up, had not been sustained in the long-term.

In the current study, dietary fat restriction was found to have both short-term and sustained benefits on blood pressure. According to our knowledge, no previous studies appear to have examined the effects of reducing dietary fat without other lifestyle interventions in people with IGT. However, modest dietary fat reduction only (37–34% after 6 months) in a group of healthy men with moderately elevated cardiovascular disease risk factors was also found to lower systolic and diastolic blood pressure by 7 and 6 mmHg, respectively [32].

The blood pressure reduction in this study was similar to the 5.4 mmHg reduction reported in a 6 year diet and physical activity intervention for diabetes prevention in people with IGT [33]. Another two diet and physical activity intervention studies, which followed participants for 1 year, reported a drop in blood pressure (4–5 mmHg) compared to control groups [34,35].

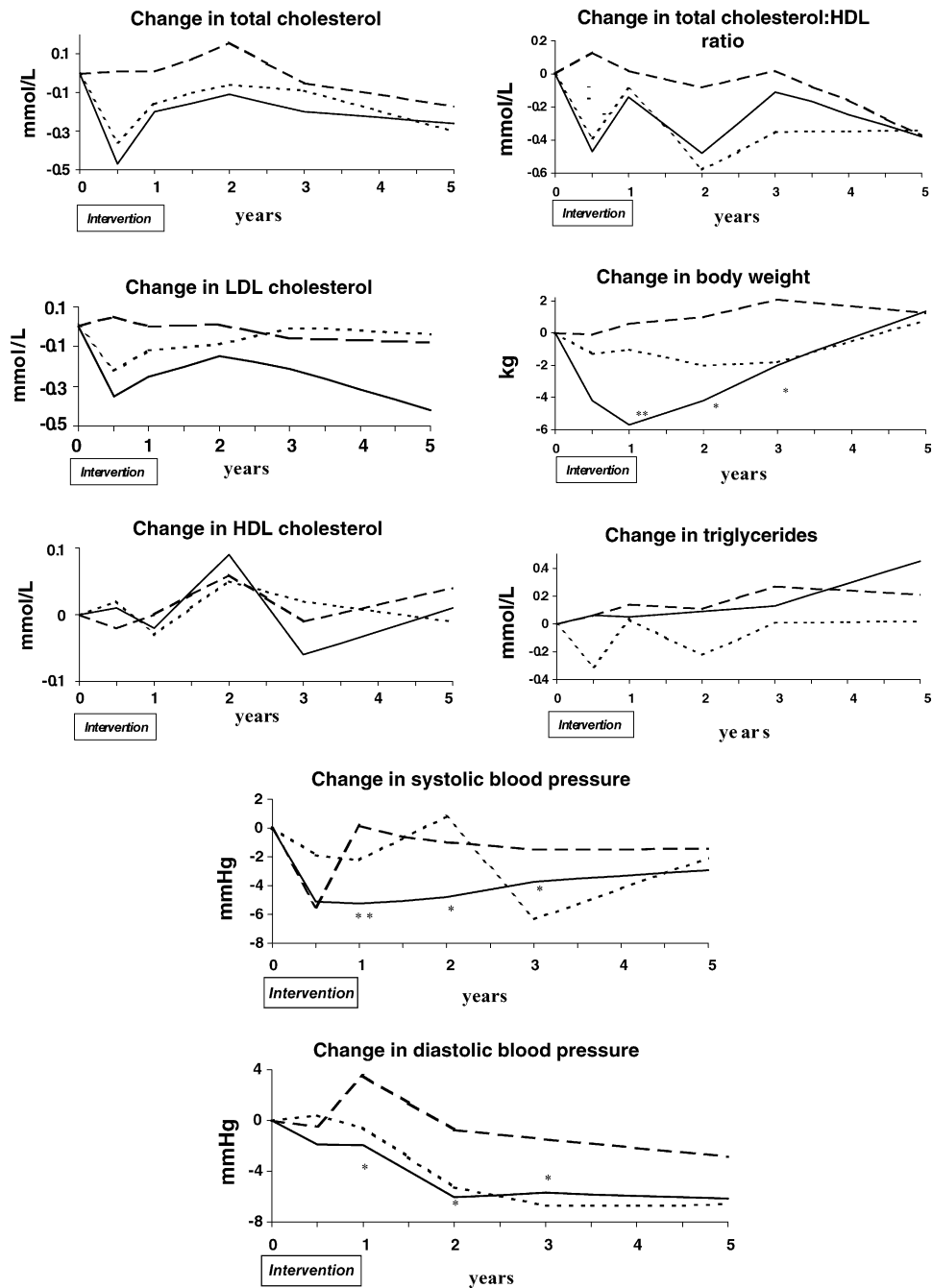


Fig. 1. Five year changes in total cholesterol, LDL cholesterol, HDL cholesterol, total cholesterol:HDL ratio, body weight, triglycerides, systolic blood pressure and diastolic blood pressure in the control diet (CD) group (— —), reduced fat 'least compliant' group (...), and reduced fat 'most compliant' group (—). Significant differences are shown between the 'most compliant' and CD groups (* $P < 0.05$, ** $P < 0.01$).

Previous studies have shown a beneficial effect of high calcium intake on blood pressure [36], thus it is possible that the intervention group were eating more calcium rich foods long-term despite showing no differences at 1 year.

Numerous studies show that the positive relationships between dietary changes and systolic blood pressure. However, this intervention had a greater effect on diastolic blood pressure than systolic blood pressure over 5 years ($P < 0.0001$ versus $P = 0.0185$). In a randomised crossover design study of a low fat, high carbohydrate diet in healthy males, subjects with the largest fall in diastolic blood pressure also experienced the largest increase in insulin sensitivity ($r = -0.697$, $P = 0.008$) [37]. Perhaps the reduced fat diet in our study, although apparently unsustained, may have had lasting effects on diastolic blood pressure long-term by improving insulin sensitivity.

However, the long-term improvements in blood pressure may not be so much a result of reduced fat eating per se, but more related to earlier reductions in body weight. Weight loss would also be expected to positively affect serum triglycerides and HDL-cholesterol. In our study, no significant differences were found for these measurements between more and less compliant reduced fat diets groups and controls. An increased intake of carbohydrate and protein relative to total energy resulted from the intervention (<0.0004 and 0.0207), but there is insufficient data to ascertain whether these changes increased foods specifically likely to affect serum lipids in a positive way and whether their consumption changed long-term.

Our study was limited by the numbers of people, particularly of Maori and Pacific Island ethnicity, lost to follow-up ($n = 24$, 17.6%). It was also limited by the manner of recruitment into the study. People who discover from a worksite survey that their blood glucose level was somewhat raised might not be as ready or willing to change lifestyle behaviours compared with people who already have known health problems such as diabetes. Therefore, overall compliance and commitment to long-term change may have been lower in these participants compared to volunteers.

It has been argued that saturated fat not total fat needs to be targeted for dietary intervention. Also some researchers suggest that advice for individuals at risk for cardiovascular disease should emphasize

maintaining moderate fat intake by replacing saturated and trans fatty acids with unsaturated oils [38,39]. This recommendation may be suitable for lean individuals, but for the overweight majority of the population a decrease in total fat will also be beneficial and will result in an overall improved cardiovascular risk profile.

This study has shown that reducing fat in people with IGT not only improves glycaemia, but also cardiovascular disease risks of the lipid profile and blood pressure. The potential adverse effects on HDL and triglyceride levels of such low fat dietary regimens appear minimal for people at risk of cardiovascular disease [40,41]. Whereas high risk individuals may benefit from a more targeted and individualised approach that addresses their specific cardiovascular risk factors, population based studies should aim to reduce overall fat intake. To ensure sustained dietary change, prolonged follow-up and environmental support are necessary, so that, low fat choices are easy choices.

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